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Research article

Equivalent orthotropic model for corrugated plates based on simplified constitutive relation

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ABSTRACT

The transverse bending stiffness of a corrugated plate determined using an equivalent orthotropic model based on the Dirichlet boundary conditions and representative volume element method is the upper limit of the real solution, albeit often inaccurate in practice. Based on the Kirchhoff hypothesis, the constitutive relations of corrugated plates were reasonably simplified to propose a new corrugated plate equivalent orthotropic model. The resulting equivalent transverse bending stiffness agreed with the results of the most reasonable simplified expression available. Static response results obtained using the proposed model were more accurate than those obtained using the existing model without a simplified constitutive relation. Furthermore, for dynamic response, the proposed model exhibited improved accuracy when predicting low- and high-order natural frequencies. For the arc-and-tangent corrugated plate, two integral parameter expressions were calculated; the proposed equivalent plate model was verified by comparing the results of a full-scale test of the corrugated metal pipe arch culvert.

1. Introduction

A corrugated plate is a lightweight structure with strong anisotropy, buckling stability, and good energy absorption capacity. It is flexible in the direction of corrugation, providing the ability to adapt to deformation, and is more rigid in the direction perpendicular to the corrugation (transverse direction), providing good bearing capacity. Corrugated structures have been widely used in the packaging industry as well as in civil, marine, and aerospace engineering [1, 2, 3, 4, 5]. The finite element analysis of corrugated plates requires a large number of precise meshes, resulting in considerable computation time. In particular, when designing and optimizing a corrugated plate structure, the considerable amount of repeated modelling and analysis required can be time-consuming and laborious. Therefore, it is necessary to establish an equivalent model to describe the waveform geometric parameters of the corrugation. When the size of a single wave is much smaller than the macro size of the corrugated panel, a common approach is to convert the corrugated panel to an orthotropic panel. Thus, the corrugated plate equivalent model has been widely used as the theoretical basis for the analysis of multilayer composite structures [6, 7].

In 1923, Huber [8] first calculated the equivalent stiffnesses of corrugated plates. In 1931, Seydel [9] followed Huber's method [8] to obtain equivalent bending stiffness formulas for a corrugated plate. In 1959, Timoshenko et al. [10] provided general bending and torsional stiffness constant expressions that considered a single corrugated steel plate shear wall (employing sinusoidal or trapezoidal wave forms) to be equivalent to an orthotropic plate. Additional studies have been conducted on a variety of corrugated plate configurations and materials using different approaches since then. McFarland [11] and Briassoulis [12] studied the equivalent bending stiffnesses of rectangular and sinusoidal corrugated plates, respectively, with the latter assuming that a corrugated plate can be analysed as an equivalent orthotropic plate with a uniform thickness. Samanta and Mukhopadhyay [13] conducted static and dynamic analyses of trapezoidal corrugated plates, simultaneously considering their tensile and bending stiffnesses. Yokozeki et al. [14] studied the stiffness characteristics of corrugated carbon epoxy resin composite laminates using experimental and analytical methods, and Dayyani et al. [15] used numerical and experimental methods to study the tensile and bending properties of corrugated laminates. Bartolozzi et al. [16] studied the load response of a sinusoidal corrugated plate through theoretical analysis and numerical simulation; Kress and

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Winkler [17] deduced accurate analytical expressions for the load response of an equivalent orthotropic plate when the corrugation is defined as a circular arc, and then, developed the planar finite element accordingly [18]; and Filipovic and Kress [19, 20] considered the lateral shear response. Mohammadi et al. [21] used Castigliano's theorem to derive analytical formulations for trapezoidal corrugated plates and verified the analysis results using tensile tests. Liew et al. [22] analysed trapezoidal and sinusoidal corrugated plates as equivalent orthogonal anisotropic plates to derive their elastic properties. Annin et al. [23] found universal relationships between the equivalent stiffnesses in different directions on corrugated plates made of homogeneous materials, including $D_{2211}^{\nu+\mu} = \nu_0 D_{2222}^{\nu+\mu}$, where subscripts 1 and 2 represent the stiffness in the transverse and corrugated direction, respectively, ν_0 stands for Poisson's ratio, and ν and μ can take the values 0 or 1.

There are two common methods for predicting the equivalent stiffnesses of periodic structures such as corrugated plates: the representative volume element method and the asymptotic homogenization method [24, 25, 26]. The representative volume element method, which has a clear mechanical concept, is a relatively simple analytical method that is very efficient in predicting the equivalent stiffnesses of complex structures and is widely applied to periodic materials. However, it is not based on a strict mathematical derivation; therefore, any resulting equivalent stiffness prediction is inherently approximate. The asymptotic homogenisation method employs the perturbation principle to predict the equivalent physical properties of a system of partial differential equations defined for a representative body element and provides a rigorous basis for mathematical theoretical derivation. However, this method can be quite complex to apply.

Hassani and Hinton [27, 28, 29, 30] first analyzed the equivalent properties of periodic structural materials using asymptotic homogenization, the principle of which is to divide the characteristics of periodic structural materials into homogeneous macroscopic features and inhomogeneous microscopic features, and then, transform these two parts into two homogenisation problems—one at macroscale and one at fine scale—through perturbation. Based on the research of Kress and Winkler [17], Xia et al. [31, 32] derived general expressions for the stiffnesses of an equivalent orthotropic plate model of a corrugated plate based on the homogenisation technique [33, 34] using Dirichlet boundary conditions and taking a single periodic waveform as a representative volume element. This equivalent model can be applied to corrugated plates with arbitrary waveforms and requires only two integrals related to their geometric parameters.

Ye et al. [35] applied the variational asymptotic method, a type of homogenization method, to obtain an equivalent orthotropic plate model of a corrugated plate that was effective for both shallow and deep corrugations. Although this model is quite accurate, its calculation is cumbersome requiring a high level of mathematical skill to correctly employ. Nevertheless, the equivalent corrugated plate stiffness formulations provided in [35] are generally considered to be the most reasonable at present. Using a method similar to the variational asymptotic method, Kolpakov and Kolpakov [36] applied two-step dimensionality reduction to reduce the three-dimensional corrugated plate problem to a one-dimensional curved beam bending problem, and thereby, obtain the equivalent stiffness formulations of the corrugated plate. This represents the simplest formulation obtained using mathematical analysis of the formulas and assumptions in [35]. Kolpakov and Kolpakov [37] then further extended the applicable scope of their proposed equivalent stiffness formulations to include any waveform (including symmetrical and asymmetrical waveforms that can and cannot be described by the function $y = y(x)$).

Recently, using the notion of structural genomes, a unified theory for multiscale constitutive modelling of composites was developed by Yu [38]. A structural genome is described as the smallest mathematical building piece of a structure, generalized from the idea of the representative volume element. Structure genome mechanics governs the information needed to bridge the microstructure length scale of composites and the macroscopic length scale of structural analysis, and it provides a unified theory for building constitutive models for structures such as three-dimensional structures, beams, plates, and shells over multiple length scales. A general-purpose computer code called SwiftComp (accessible at https://cdmhub.org/resources/scstandard) implements the unified theory and is used in a few example cases to demonstrate its application. Yu [39] proposed a simplified description of the mechanics of structure genome (MSG). It generalizes prior research efforts based on the variational asymptotic technique (VAM), such as variational asymptotic beam section analysis (VABS), variational asymptotic plate and shell analysis (VAPAS), and variational asymptotic approach for unit cell homogenization. The similarities and differences between MSG and prior works are clearly stated. Deo and Yu [40] Using structure genome mechanics, equivalent plate properties were obtained for corrugated structures. The proposed theory has been validated by incorporating it into a computer code known as MSG-TW, which stands for MSG thin-walled model. Comparisons with prior work have been made to demonstrate the benefits of the new model over existing ones.

Ye et al. [35] believed that the correct stiffnesses were obtained in reference [31], except for the determination of the transverse bending stiffness, which remains to be revised. Notably, previous studies [41, 42] have determined that, when using the representative volume element method, the results obtained using the Dirichlet boundary condition were similar to those obtained using the Voigt upper bound [43], whereas the results obtained using the Neumann boundary condition were similar to those obtained using the Reuss lower bound [44]. Furthermore, Aoki and Maysenhölder [45] studied the natural frequencies of symmetric sinusoidal and trapezoidal corrugated free rectangular plates through experiments and finite element analyses of an equivalent plate model using the transverse bending stiffness expression of Ye et al. [35] and the torsional stiffness expression of Xia et al. [31], as well as other undisputed stiffness expressions. However, they found that the natural frequencies predicted by the selected equivalent plate model were only satisfactory for low-order modes.

Arc-and-tangent corrugated plates are widely used in buried corrugated metal pipe structures. When performing 3D finite element analysis, the equivalent orthotropic plate methods commonly used in this field belong to the engineering method [46, 47]. These methods are based on the equivalent bending stiffness (or moment of inertia of the section) and the equivalent plate cross-sectional area (or tensile stiffness) and used to obtain the equivalent plate thickness (usually dozens of times the original plate thickness). Subsequently, other elastic parameters, such as equivalent transverse elastic modulus and longitudinal elastic modulus ([46] based on the longitudinal section moment of inertia is equal [47]; no display analytical solution is provided), and other elastic parameters are calculated. The section parameters in these methods do not provide analytical expressions and usually refer to the values in the specifications (or those acquired by AutoCAD).

Therefore, in this study, the classical Kirchhoff hypothesis was applied to reasonably simplify the constitutive relation determined according to the Dirichlet boundary condition—that is, the generalized strain boundary condition— in a representative volume element method-based model under the theoretical framework presented in [31], to derive simplified expressions for the equivalent stiffnesses. Compared with other more complex formulations in the extant literature, the expression for equivalent transverse bending stiffness in this study is consistent with the simplified result of the most reasonable known expression presented in [35]. Taking two typical types of corrugated plate as examples, the accuracy of the static response predicted by the proposed equivalent stiffness model under different boundary conditions was found to be superior to that obtained by previously proposed models. The dynamic vibration characteristics of a corrugated plate with four simply supported edges were also evaluated with the proposed equivalent stiffness model; the resulting natural frequencies were more accurate than those obtained by the Xia et al. model [31]. The parameters related to the equivalent model of the arc-and-tangent

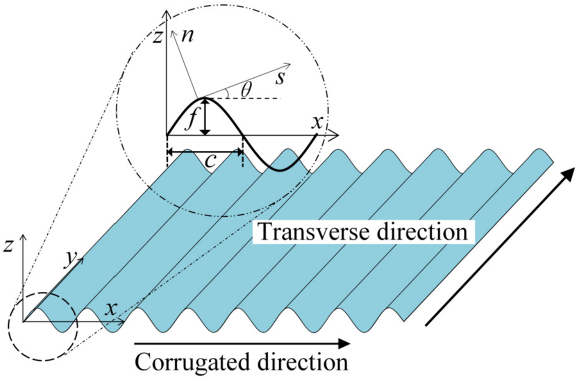

Fig. 1. Corrugated plate coordinate system.

corrugated plates in engineering were calculated, and the section characteristic parameters calculated based on these parameters were in good agreement with the specification values. By comparing the results of a full-scale test of the corrugated metal pipe arch culvert, it was revealed that the results of the proposed equivalent model were generally slightly better than those of the engineering method.

2. Equivalent corrugated plate model

2.1. Overview of Xia's equivalent corrugated plate model

A corrugated plate made of an isotropic material with an elastic modulus E and Poisson's ratio υ is taken as an example for the theoretical derivation in this study. The Cartesian coordinate system and local curve coordinate system established at the middle plane of the corrugated plate are consistent with those of Xia et al. [31], as shown in Fig. 1. Here, f represents the half-height of the corrugated plate in the z direction, and c represents the half-period of a single corrugation.

In this coordinate system, the components of the coupling stiffness matrix in the constitutive relation are zero, and the constitutive relation of the corrugated plate is approximately equivalent to that of an orthotropic plate as follows:

$$\begin{Bmatrix} \overline{N}_x \\ \overline{N}_y \\ \overline{N}_{xy} \\ \overline{M}_x \\ \overline{M}_y \\ \overline{M}_{xy} \end{Bmatrix} = \begin{bmatrix} \overline{A}_{11} & \overline{A}_{12} & 0 & 0 & 0 & 0 \\ \overline{A}_{12} & \overline{A}_{22} & 0 & 0 & 0 & 0 \\ 0 & 0 & \overline{A}_{66} & 0 & 0 & 0 \\ 0 & 0 & 0 & \overline{D}_{11} & \overline{D}_{12} & 0 \\ 0 & 0 & 0 & \overline{D}_{12} & \overline{D}_{22} & 0 \\ 0 & 0 & 0 & 0 & 0 & \overline{D}_{66} \end{bmatrix} \begin{Bmatrix} \overline{\varepsilon}_x \\ \overline{\varepsilon}_y \\ \overline{\varepsilon}_{xy} \\ \overline{\kappa}_x \\ \overline{\kappa}_y \\ \overline{\kappa}_{xy} \end{Bmatrix}, \tag{1}$$

where $\overline{\varepsilon}_x$, $\overline{\varepsilon}_y$, $\overline{\varepsilon}_{xy}$ and $\overline{\kappa}_x$, $\overline{\kappa}_y$, $\overline{\kappa}_{xy}$ represent the strain and curvature components, respectively, of the middle plane in the orthotropic plate model; and $\overline{N}_x$, $\overline{N}_y$, $\overline{N}_{xy}$ and $\overline{M}_x$, $\overline{M}_y$, $\overline{M}_{xy}$ represent the force and moment components, respectively, per unit width. Eq. (1) is rewritten into the equivalent compact form as follows:

$$\begin{Bmatrix} \overline{N} \\ \overline{M} \end{Bmatrix} = \begin{bmatrix} \overline{A} & \mathbf{0} \\ \mathbf{0} & \overline{D} \end{bmatrix} \begin{Bmatrix} \overline{\varepsilon} \\ \overline{\kappa} \end{Bmatrix} \tag{2}$$

where the definition of the vectors and matrices can be understood by comparing Eqs. (1) and (2).

As the corrugated plate is generated by repeating the base element on a regular basis, the stiffness properties of the equivalent plate may be estimated using a base element known as a representative volume element. In the local curvilinear coordinates, the constitutive relation of the corrugated plate is as follows:

$$\begin{Bmatrix} N_s \\ N_y \\ N_{sy} \\ M_s \\ M_y \\ M_{sy} \end{Bmatrix} = \begin{bmatrix} A_{11} & A_{12} & 0 & 0 & 0 & 0 \\ A_{12} & A_{22} & 0 & 0 & 0 & 0 \\ 0 & 0 & A_{66} & 0 & 0 & 0 \\ 0 & 0 & 0 & D_{11} & D_{12} & 0 \\ 0 & 0 & 0 & D_{12} & D_{22} & 0 \\ 0 & 0 & 0 & 0 & 0 & D_{66} \end{bmatrix} \begin{Bmatrix} \varepsilon_s \\ \varepsilon_y \\ \varepsilon_{sy} \\ \varepsilon_s \\ \varepsilon_y \\ \varepsilon_{sy} \end{Bmatrix}, \tag{3}$$

where,

$$A_{11} = \frac{Eh}{1-v^2}, \qquad A_{12} = \frac{vEh}{1-v^2},$$
$$A_{22} = \frac{Eh}{1-v^2}, \qquad A_{66} = \frac{Eh}{2(1+v)},$$
$$D_{11} = \frac{Eh^3}{12(1-v^2)}, \qquad D_{12} = \frac{vEh^3}{12(1-v^2)},$$
$$D_{22} = \frac{Eh^3}{12(1-v^2)}, \qquad D_{66} = \frac{Eh^3}{24(1+v)}.$$

The terms of Eq. (1) are evaluated by the equivalent energy method and equal effectiveness method. The strain energy of the original corrugated panel is determined using the equivalent energy method as

$$U = \frac{1}{2}\iint \mathbf{N}^{\mathrm{T}}\mathbf{S}\mathbf{N}\,ds\,dy, \tag{4}$$

where $\mathbf{N} = \left[N_s, N_y, N_{sy}, M_s, M_y, M_{sy}\right]^{\mathrm{T}}$. The original plain composite sheet material's flexibility matrix, $\mathbf{S}$, is the inverse of the stiffness matrix, $\mathbf{K}$ (as in Eq. (3)), and is given by

$$\mathbf{S} = \mathbf{K}^{-1} = \begin{bmatrix} \frac{A_{22}}{A_{11}A_{22}-A_{12}^2} & \frac{-A_{12}}{A_{11}A_{22}-A_{12}^2} & 0 & 0 & 0 & 0 \\ \frac{-A_{12}}{A_{11}A_{22}-A_{12}^2} & \frac{A_{11}}{A_{11}A_{22}-A_{12}^2} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{A_{66}} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{D_{22}}{D_{11}D_{22}-D_{12}^2} & \frac{-D_{12}}{D_{11}D_{22}-D_{12}^2} & 0 \\ 0 & 0 & 0 & \frac{-D_{12}}{D_{11}D_{22}-D_{12}^2} & \frac{D_{11}}{D_{11}D_{22}-D_{12}^2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{D_{66}} \end{bmatrix}, \tag{5}$$

Eq. (4) integrates across a unit cell whose length corresponds to one cycle of the corrugation and breadth is b. The strain energy U can be obtained by substituting Eq. (5) with Eq. (4).

The strain energy $\overline{U}$ of the equivalent orthotropic plate is identical to the strain energy U of the equivalent orthotropic plate, according to the equivalent energy technique. Therefore,

$$U = \overline{U} = \frac{1}{2}(2c)b \begin{Bmatrix} \overline{\varepsilon} \\ \overline{\kappa} \end{Bmatrix}^{\mathrm{T}} \begin{bmatrix} \overline{A} & \mathbf{0} \\ \mathbf{0} & \overline{D} \end{bmatrix} \begin{Bmatrix} \overline{\varepsilon} \\ \overline{\kappa} \end{Bmatrix}, \tag{6}$$

where $2c$ is the corrugation period, and b is the breadth of the unit cell. The stiffness components are then determined by successively applying different strain boundary conditions. Eq. (6) is used to calculate the stiffness components of the equivalent orthotropic plate.

For the six distinct strain boundary conditions presented in Table 1, the equivalent force method compares the equivalent internal forces and moments to the average internal forces and moments of the original corrugated representative volume element. A suitable choice of comparable internal force and required strain boundary condition yields different stiffness components. Table 1 includes detailed information. A more detailed description of the stiffness matrix derivation can be found in Section 2.3 of [31].

Finally, [31] presented new corrugated panel equivalent models. The generic general equivalent stiffness component expressions for the corrugated plate described by $(x(s), z(s))$ are summarized as follows:

$$\overline{A}_{11} = \frac{2c}{\left[\frac{I_1}{A_{11}} + \frac{I_2}{D_{11}}\right]} = \frac{1}{1-v^2}\frac{2cEh^3}{I_1h^2 + 12I_2},$$

$$\overline{A}_{12} = \frac{A_{12}}{A_{11}}\overline{A}_{11} = \frac{v}{1-v^2}\frac{2cEh^3}{I_1h^2 + 12I_2},$$

Table 1. Boundary conditions of equivalent orthotropic plates and corresponding stiffness expressions according to Xia et al. [31].

Boundary conditions $\begin{Bmatrix} \bar{\varepsilon} \\ \bar{\kappa} \end{Bmatrix}$	Equivalent energy method	Equivalent force method
$\{1,0,0,0,0,0\}^T$	$\overline{A}_{11} = 2U/(2cb)$	$\overline{A}_{11} = \overline{N}_x$; $\overline{A}_{12} = \overline{N}_y$
$\{0,1,0,0,0,0\}^T$	$\overline{A}_{22} = 2U/(2cb)$	$\overline{A}_{22} = \overline{N}_y$; $\overline{A}_{12} = \overline{N}_x$
$\{0,0,1,0,0,0\}^T$	$\overline{A}_{66} = 2U/(2cb)$	$\overline{A}_{66} = \overline{N}_{xy}$
$\{0,0,0,1,0,0\}^T$	$\overline{D}_{11} = 2U/(2cb)$	$\overline{D}_{11} = \overline{M}_x$; $\overline{D}_{12} = \overline{M}_y$
$\{0,0,0,0,1,0\}^T$	$\overline{D}_{22} = 2U/(2cb)$	$\overline{D}_{22} = \overline{M}_y$; $\overline{D}_{12} = \overline{M}_x$
$\{0,0,0,0,0,1\}^T$	$\overline{D}_{66} = 2U/(2cb)$	$\overline{D}_{66} = \overline{M}_{xy}$

$$
\begin{aligned}
&\overline{A}_{22} = \frac{\overline{A}_{12}A_{12}}{A_{11}} + \frac{l}{c}\frac{A_{11}A_{22} - A_{12}^2}{A_{11}} = \frac{\nu^2}{1-\nu^2}\frac{2cEh^3}{I_1h^2 + 12I_2} + \frac{lEh}{c},\\
&\overline{A}_{66} = \frac{c}{l}A_{66} = \frac{cEh}{2l(1+\nu)},\\
&\overline{D}_{11} = \frac{c}{l}D_{11} = \frac{1}{(1-\nu^2)}\frac{cEh^3}{12l},\\
&\overline{D}_{12} = \frac{D_{12}}{D_{11}}\overline{D}_{11} = \frac{\nu}{(1-\nu^2)}\frac{cEh^3}{12l},\\
&\overline{D}_{22} = \frac{1}{2c}[I_2A_{22} + I_1D_{22}] = \frac{1}{(1-\nu^2)}\frac{12I_2Eh + I_1Eh^3}{24c},\\
&\overline{D}_{66} = \frac{l}{c}D_{66} = \frac{1}{(1+\nu)}\frac{lEh^3}{24c}.
\end{aligned} \tag{7}
$$

Eq. (7) shows that the constants that must be computed for a particular corrugated geometry are the half-period c, expanded half-length of the corrugation wave l, and integrals I_1 and I_2.

2.2. Simplification of constitutive relation

As previously stated, prior works [38, 39] found that when employing the representative volume element technique, findings obtained using Dirichlet boundary conditions are identical to those obtained using Voigt upper bound [43]. In this section, the Kirchhoff hypothesis is applied to the generalized strain boundary conditions, yielding several new stiffness component expressions of the original corrugated plate constitutive equation. This method may reduce the influence of Dirichlet boundary conditions and produce stiffness estimates closer to the actual stiffness. These new stiffness component expressions shall be utilized to assess equivalent orthotropic plate stiffness component expressions.

In the local coordinate system employed in this study, the Kirchhoff thin plate calculation hypothesis is used based on the following assumptions:

1) The positive strain ε_n perpendicular to the midplane of the plate is ignored.

2) The stress components τ_{ns}, τ_{yn}, and σ_n are much smaller than the other three stress components (τ_{sy}, σ_s, and σ_y), so they are considered secondary, and the strain they cause can be ignored (note, however, that these stress components are necessary to maintain balance and cannot be ignored).

According to these two assumptions, the constitutive relation of the completely elastic three-dimensional isotropic body in the local coordinate system, namely, the generalised Hooke's law, can be simplified from six equations to three equations as follows:

$$
\left.\begin{aligned}
&\varepsilon_s = \tfrac{1}{E}\left[\sigma_s - \nu\left(\sigma_y + \sigma_n\right)\right]\\
&\varepsilon_y = \tfrac{1}{E}\left[\sigma_y - \nu\left(\sigma_n + \sigma_s\right)\right]\\
&\varepsilon_n = \tfrac{1}{E}\left[\sigma_n - \nu\left(\sigma_s + \sigma_y\right)\right]\\
&\gamma_{yn} = \tfrac{1}{G}\tau_{yn}\\
&\gamma_{ns} = \tfrac{1}{G}\tau_{ns}\\
&\gamma_{sy} = \tfrac{1}{G}\tau_{sy}
\end{aligned}\right\} \Rightarrow \left.\begin{aligned}
&\varepsilon_s = \tfrac{1}{E}\left(\sigma_s - \nu\sigma_y\right)\\
&\varepsilon_y = \tfrac{1}{E}\left(\sigma_y - \nu\sigma_s\right)\\
&\gamma_{sy} = \tfrac{1}{G}\tau_{sy}
\end{aligned}\right\}. \tag{8}
$$

2.2.1. Generalised strain boundary condition $\{1,0,0,0,0,0\}^T$

According to Table 1, the symmetry of the corrugated plate with the generalised strain boundary condition $\{1,0,0,0,0,0\}^T$ requires that most local strains be zero; thus, $\varepsilon_y = 0$, $\gamma_{sy} = 0$, $\kappa_y = 0$, and $\kappa_{sy} = 0$.

Since ε_y is equal to 0, we can apply the two Kirchhoff assumptions stated above to ignore the normal strain ε_y and the strain caused by the stress components σ_y and τ_{sy}.

The Eq. (8) in the local coordinate can then be simplified as

$$
\left.\begin{aligned}
&\varepsilon_s = \tfrac{1}{E}\left(\sigma_s - \nu\sigma_y\right)\\
&\varepsilon_y = \tfrac{1}{E}\left(\sigma_y - \nu\sigma_s\right)\\
&\gamma_{sy} = \tfrac{1}{G}\tau_{sy}
\end{aligned}\right\} \Rightarrow \varepsilon_s = \frac{1}{E}\sigma_s = \frac{N_s}{Eh}. \tag{9}
$$

This state (as in Eq. (9)) is like Hooke's law under uniaxial stress, where $\tilde{A}_{11} = Eh$ and $\tilde{A}_{12} = \nu\tilde{A}_{11}$ remain the same; thus, $\tilde{A}_{12} = \nu Eh$.

2.2.2. Generalised strain boundary condition $\{0,1,0,0,0,0\}^T$

Under the generalised strain boundary condition $\{0,1,0,0,0,0\}^T$, $\varepsilon_y = 1$, $\gamma_{sy} = 0$, $\kappa_y = 0$, and $\kappa_{sy} = 0$. As ε_s can be considered negligible, assume $\varepsilon_s = 0$. The three physical equations in the local coordinate can then be reduced to Hooke's law under uniaxial stress as follows:

$$
\left.\begin{aligned}
&\varepsilon_s = \tfrac{1}{E}\left(\sigma_s - \nu\sigma_y\right)\\
&\varepsilon_y = \tfrac{1}{E}\left(\sigma_y - \nu\sigma_s\right)\\
&\gamma_{sy} = \tfrac{1}{G}\tau_{sy}
\end{aligned}\right\} \Rightarrow \varepsilon_y = \frac{1}{E}\sigma_y = \frac{N_y}{Eh}. \tag{10}
$$

In this case (as in Eq. (10)), $\tilde{A}_{22} = Eh$.

2.2.3. Generalised strain boundary condition $\{0,0,0,0,1,0\}^T$

For the $\{0,0,0,0,1,0\}^T$ generalised strain boundary condition, the strain component in the local coordinate system of the corrugated plate is defined as

$$
\left\{\varepsilon_s, \varepsilon_y, \gamma_{sy}, \kappa_s, \kappa_y, \kappa_{sy}\right\} = \left\{0, z, 0, \frac{dx}{ds}, 0, 0\right\}. \tag{11}
$$

Under the conditions expressed in Eq. (11), by applying Kirchhoff's first two assumptions as stated above, a simplified physical equation can be obtained:

$$
M_y = \frac{Eh^3}{12}\kappa_y. \tag{12}
$$

In this case (as in Eq. (12)), $\tilde{D}_{22} = \frac{Eh^3}{12}$.

Substituting $\tilde{A}_{11}$, $\tilde{A}_{12}$, $\tilde{A}_{22}$, and $\tilde{D}_{22}$ for A_{11}, A_{12}, A_{22}, and D_{22} in Eq. (3) provides the Eq. (13). In summary, the constitutive relation of a corrugated plate is simplified as follows:

$$
\begin{Bmatrix} N_s \\ N_y \\ N_{sy} \\ M_s \\ M_y \\ M_{sy} \end{Bmatrix} = \begin{bmatrix} Eh & \nu Eh & 0 & 0 & 0 & 0 \\ \nu Eh & Eh & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{Eh}{2(1+\nu)} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{Eh^3}{12(1-\nu^2)} & \frac{\nu Eh^3}{12(1-\nu^2)} & 0 \\ 0 & 0 & 0 & \frac{\nu Eh^3}{12(1-\nu^2)} & \frac{Eh^3}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{Eh^3}{24(1+\nu)} \end{bmatrix} \begin{Bmatrix} \varepsilon_s \\ \varepsilon_y \\ \gamma_{sy} \\ \kappa_s \\ \kappa_y \\ \kappa_{sy} \end{Bmatrix}. \tag{13}
$$

2.3. Equivalent universal stiffness model

Using the simplified constitutive relation Eq. (13), the stiffness matrix components of the equivalent orthogonal anisotropic plate are derived using the equivalent energy approach and equivalent force method, as in [31]. Section 2.3 of [31] contains a more detailed description of the stiffness matrix derivation. To facilitate comparative analysis, the equivalent stiffness expressions of [31, 35] are expressed in symbols in this paper, as summarized in Table 2. As the general

Table 2. Equivalent stiffness expressions of corrugated panels equivalent model.

Equivalent stiffness	Equivalent model from [31] Concrete expression	Equivalent model derived in this study Concrete expression	Equivalent model from [35] Concrete expression	
			For symmetric corrugations	For shallow corrugations
$\overline{A}_{11}$	$\frac{1}{1-\nu^2}\frac{2cEh^3}{I_1h^2+12I_2}$	$\frac{2cEh^3}{I_1h^2+12I_2(1-\nu^2)}$	$\frac{1}{(1-\nu^2)}\frac{2cEh^3}{12I_2}$	$\frac{1}{1-\nu^2}\frac{I_1Eh}{2c}$
$\overline{A}_{12}$	$\frac{\nu}{1-\nu^2}\frac{2cEh^3}{I_1h^2+12I_2}$	$\frac{2c\nu Eh^3}{I_1h^2+12I_2(1-\nu^2)}$	$\frac{\nu}{(1-\nu^2)}\frac{2cEh^3}{12I_2}$	$\frac{\nu}{1-\nu^2}\frac{I_1Eh}{2c}$
$\overline{A}_{22}$	$\frac{\nu^2}{1-\nu^2}\frac{2cEh^3}{I_1h^2+12I_2}+\frac{lEh}{c}$	$\frac{2cE\nu^2h^3}{I_1h^2+12I_2(1-\nu^2)}+\frac{(1-\nu^2)lEh}{c}$	$\frac{Ehl}{c}$	$\frac{\nu^2}{1-\nu^2}\frac{I_1Eh}{2c}+\frac{lEh}{c}$
$\overline{A}_{66}$	$\frac{cEh}{2l(1+\nu)}$	$\frac{cEh}{2l(1+\nu)}$	$\frac{cEh}{2l(1+\nu)}$	$\frac{cEh}{2l(1+\nu)}$
$\overline{D}_{11}$	$\frac{1}{(1-\nu^2)}\frac{cEh^3}{12l}$	$\frac{1}{(1-\nu^2)}\frac{cEh^3}{12l}$	$\frac{1}{(1-\nu^2)}\frac{cEh^3}{12l}$	$\frac{1}{(1-\nu^2)}\frac{cEh^3}{12l}$
$\overline{D}_{12}$	$\frac{\nu}{(1-\nu^2)}\frac{cEh^3}{12l}$	$\frac{\nu}{(1-\nu^2)}\frac{cEh^3}{12l}$	$\frac{\nu}{(1-\nu^2)}\frac{cEh^3}{12l}$	$\frac{\nu}{(1-\nu^2)}\frac{cEh^3}{12l}$
$\overline{D}_{22}$	$\frac{1}{(1-\nu^2)}\frac{12I_2Eh+I_1Eh^3}{24c}$	$\frac{12I_2Eh+I_1Eh^3}{24c}$	$\frac{I_2Eh}{2c}$	$\frac{I_1Eh^3}{24c}+\frac{\nu^2}{1-\nu^2}\frac{cEh^3}{12l}$
$\overline{D}_{66}$	$\frac{1}{(1+\nu)}\frac{lEh^3}{24c}$	$\frac{1}{(1+\nu)}\frac{lEh^3}{24c}$	$\frac{1}{(1+\nu)}\frac{lEh^3}{24c}$	$\frac{1}{(1+\nu)}\frac{lEh^3}{24c}$

where $I_1=\int_0^{2l}\left(\frac{\mathrm{d}x}{\mathrm{d}s}\right)^2\mathrm{d}s$, $I_2=\int_0^{2l}z^2\mathrm{d}s$.

Table 3. Various expressions for equivalent transverse bending stiffness.

Reference	Original expression	Expression using symbols in this paper
[36]	$\nu^2\frac{Eh^3}{12(1-\nu^2)}\frac{p}{L}+\frac{E}{P}\left[h\int_0^L x_3(s)^2ds+\frac{h^3}{12}\int_0^L\cos^2\varphi(s)ds\right]$	$\nu^2\overline{D}_{11}+\frac{I_2Eh}{2c}+\frac{I_1Eh^3}{24c}$
[37]	$\nu^2\frac{Eh^3}{12(1-\nu^2)}\frac{p}{L}+\frac{EhL}{p}\left[\frac{h^2}{12}\left\langle\cos^2\varphi\right\rangle_s+\left\langle x_3^2\right\rangle_s\right]$	$\nu^2\overline{D}_{11}+\frac{I_2Eh}{2c}+\frac{I_1Eh^3}{24c}$
[35]	$\nu^2\frac{Eh^3}{12(1-\nu^2)}\frac{1}{\left\langle\sqrt{a}\right\rangle}+Eh\varepsilon^2\left\langle\phi^2\sqrt{a}\right\rangle+\frac{Eh^3}{12}\left\langle\frac{1}{\sqrt{a}}\right\rangle$	$\nu^2\overline{D}_{11}+\frac{I_2Eh}{2c}+\frac{I_1Eh^3}{24c}$
[31]	$\frac{1}{2c}[I_2A_{22}+I_1D_{22}]$	$\frac{1}{(1-\nu^2)}\left(\frac{I_2Eh}{2c}+\frac{I_1Eh^3}{24c}\right)$
Present study	-	$\frac{I_2Eh}{2c}+\frac{I_1Eh^3}{24c}$

equivalent stiffness expression [35] involves multiple parameters and has complex forms, it is difficult to express it using the symbols in this paper. Nevertheless, [35] provides the equivalent stiffness expressions of shallow waves and symmetric waves, which can be expressed using the symbols in this paper, as shown in Table 2.

The expressions $\overline{A}_{11}$, $\overline{A}_{12}$, $\overline{A}_{22}$ and $\overline{D}_{22}$ are different. For shallow corrugations and symmetric corrugations, stiffness expressions are reasonably simplified in [35], so it is normal for the two expressions in [35] to differ from each other. Ye et al. [35] believed that the correct stiffnesses were obtained in [31], except for the determination of the transverse bending stiffness. Due to the differences in expressions, the next section compares the equivalent model proposed in this paper with previous work.

3. Static verification of equivalent model

3.1. Comparison of equivalent transverse bending stiffness expressions

Ye et al. [35] believed that the correct stiffness expressions were provided in [31], except for the calculation of equivalent transverse bending stiffness. In this section, the formulas for the equivalent transverse bending stiffness ($\overline{D}_{22}$) proposed in different studies are compared with their analysed differences and commonalities. For convenience, the symbols are adopted uniformly in this study, and the expressions presented in different papers are rewritten accordingly, as shown in Table 3.

As can be observed in Table 3, the expressions from [35, 36, 37] are identical. Both the first and third terms of these expressions include h^3, while the second term of these expressions provides h. Considering that only the third term contains the geometric features of the waveform, the expression derived in this paper can be obtained by retaining the third term and ignoring the first term of the formulas from [35, 36, 37]. Compared with the expression derived in this paper, the equivalent transverse bending stiffness formula from [31] contains the additional coefficient $1/(1-\nu^2)$. For isotropic materials, $\nu<0.5$; thus, this coefficient is consistently greater than 1, resulting in a larger equivalent transverse bending stiffness than that obtained using the other formulas. This finding verifies that the equivalent transverse bending stiffness obtained using the Dirichlet boundary condition in [31] is the upper limit of the true solution and that it is reasonable to simplify the constitutive relation in this study to correct this overestimation.

Table 4. Typical corrugated plate parameters.

Corrugation	E (GPa)	ν	ρ (kg/m^3)	c (m)	f (m)	h (m)	α (°)
Sinusoidal	30	0.2	7830	0.32	0.11	0.005	-
Trapezoidal	21	0.3	7850	0.0508	0.0127	0.00635	45

3.2. Evaluation of proposed equivalent model

To verify the accuracy of the static response predictions obtained using the proposed equivalent model, two types of corrugated plates with typical sinusoidal and trapezoidal waveforms were investigated. Note that as the two integrals in the proposed equivalent stiffness expressions are related to the specific waveform, numerical solutions are required when explicit solutions are not available. For the sinusoidal corrugated plate considered in this study, the relevant physical quantities and integral expressions are as follows:

$$z(x)=f\sin\left(\frac{\pi x}{c}\right),\quad l=\int_0^c\sqrt{1+z'^2}dx,\quad I_1=\int_0^{2c}\cos^2\theta\sqrt{1+z'^2}dx,$$
$$I_2=\int_0^{2c}z^2\sqrt{1+z'^2}dx \tag{14}$$

For the trapezoidal corrugated plate considered in this study, the relevant physical quantities and integral expressions are as follows:

$$l=\frac{2f}{\sin\alpha}+c-\frac{2f}{\tan\alpha},\quad I_1=\frac{4f\cos^2\alpha}{\sin\alpha}+2c-\frac{4f}{\tan\alpha},$$
$$I_2=\frac{4f^3}{3\sin\alpha}+2f^2\left(c-\frac{2f}{\tan\alpha}\right) \tag{15}$$

For the convenience of discussion, Table 4 lists the material parameters, and Figs. 2a and b show the geometric characteristics of the sinusoidal corrugated plate and trapezoidal corrugated plate, respectively.

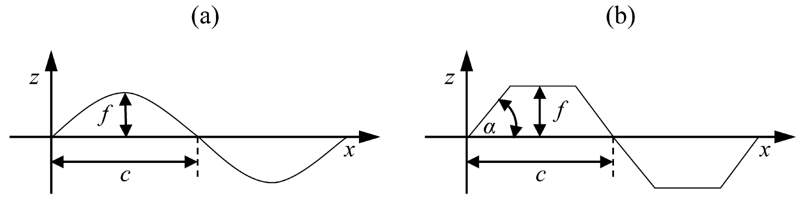

Fig. 2. Geometry of (a) sinusoid corrugated plate and (b) trapezoidal corrugated plate.

Table 5. Equivalent stiffness of sinusoidal corrugated plate.

	Briassoulis [10]	Xia et al. [31]	VAPAS [48]	Ye et al. [35]	Present study
$\overline{A}_{11}$ (N/m)	39639	47613	48152	47613	47612
$\overline{A}_{12}$ (N/m)	7928	9523	9630	9523	9522
$\overline{A}_{22}$ (MN/m)	187.08	187.08	186.92	187.08	179.60
$\overline{A}_{66}$ (MN/m)	62.500	50.113	50.097	50.113	50.113
$\overline{D}_{11}$ (N·m)	261.004	261.004	263.972	261.004	261.004
$\overline{D}_{12}$ (N·m)	52.20	52.20	52.95	52.20	52.20
$\overline{D}_{22}$ (N·m)	907830	1068260	1022874	1025540	1025530
$\overline{D}_{66}$ (N·m)	260.42	162.39	163.38	162.39	162.39

Table 6. Equivalent stiffness of trapezoidal corrugated plate.

	Samanta and Mukhopadhyay [13]	MSG-TW [40]	Xia et al. [31]	Present study
$\overline{A}_{11}$ (MN/m)	4.150	4.052	4.052	4.042
$\overline{A}_{12}$ (MN/m)	1.245	1.216	1.216	1.213
$\overline{A}_{22}$ (MN/m)	176.888	161.332	161.332	146.844
$\overline{A}_{66}$ (MN/m)	42.489	42.489	42.489	42.489
$\overline{D}_{11}$ (N·m)	371.205	407.913	407.917	407.917
$\overline{D}_{12}$ (N·m)	0	122.375	122.375	122.375
$\overline{D}_{22}$ (N·m)	31647	16242	17809	16206
$\overline{D}_{66}$ (N·m)	208.032	208.033	208.032	208.032

3.2.1. Sinusoidal corrugated plate

For the sinusoidal corrugated plate, the equivalent stiffnesses were calculated according to the equivalent orthotropic plate models presented by different scholars and proposed in this study; According to the parameters in Table 4, Eq. (14) can be used to calculate the equivalent stiffness of [31] and this paper, respectively. The results are compared in Table 5.

The equivalent plate model presented by Ye et al. [35] provided high precision, with results close to those calculated by VAPAS, which was proposed by Lee and Yu [48] for the equivalent plate modelling of microstructural plates based on the original three-dimensional elastic theory, and as such, was considered to be the 'correct' solution in this comparison. The results calculated by the equivalent model proposed in this study were almost similar to those provided by Ye et al. [35], except that $\overline{A}_{22}$ was 3.7% smaller.

3.2.2. Trapezoidal corrugated plate

For a trapezoidal corrugated plate, the equivalent stiffnesses were calculated according to the equivalent orthotropic plate models presented by different scholars and proposed in this study; According to the parameters in Table 4, Eq. (15) can be used to calculate the equivalent stiffness of [31] and this paper, respectively. The results are compared in Table 6. In both directions, the stiffness properties of the model proposed by the present study are quite identical to MSG-TW [40] values. Although the accuracy of the results of this study is not high enough in the in-plane stiffness, it is still feasible to use the equivalent bending stiffness to calculate the elastic constant of orthotropic plates in engineering.

To provide a basis for the determination of accuracy, the ABAQUS finite element analysis software was employed to simulate the $\overline{D}_{22}$ response of the trapezoidal corrugated plate using S4R shell elements. Multiple parameters must be set in the material properties of the orthotropic plates to do so. The calculation of the elastic parameters under the condition of out-of-plane loading of the equivalent model is as follows:

$$E_x = \frac{12\left(\overline{D}_{11}\overline{D}_{22} - \overline{D}_{12}^2\right)}{h^3\overline{D}_{22}}, \quad E_y = \frac{12\left(\overline{D}_{11}\overline{D}_{22} - \overline{D}_{12}^2\right)}{h^3\overline{D}_{11}}, \quad \nu_{xy} = \frac{\overline{D}_{12}}{\overline{D}_{22}},$$
$$G_{xy} = \frac{12\overline{D}_{66}}{h^3}, \quad G_{zx} = \frac{E_x}{2(1+\nu)}, \quad G_{yz} = \frac{E_y}{2(1+\nu_{xy})}. \tag{16}$$

The density of the equivalent orthotropic plate was calculated according to the equal mass of the plate as follows:

$$\rho_{eq} 2cbh = \rho 2lbh \Rightarrow \rho_{eq} = \rho\frac{l}{c}. \tag{17}$$

The exact solution for calculating the bending deflection of a rectangular anisotropic plate with four simply supported edges under uniform pressure can be found in Jones [49]:

$$w(x,y) = \frac{16p_0}{\pi^6}\sum_{m=1}^{\infty}\sum_{n=1}^{\infty}\frac{1}{mn}\frac{\sin(m\pi\frac{x}{a})\sin(n\pi\frac{y}{b})}{D_1(\frac{m}{a})^4 + 2H(\frac{mn}{ab})^2 + D_2(\frac{n}{b})^4}, \tag{18}$$

where $D_1 = \overline{D}_{11}$, $D_2 = \overline{D}_{22}$, and $H = \overline{D}_{12} + 2\overline{D}_{66}$; p_0 is the uniform pressure in N/m^2; a and b are the lengths of the corrugated plate in the corrugated direction and transverse direction, respectively; and m and n are positive integers representing the half sine-wave numbers in the x and y directions, respectively.

A finite element model of a trapezoidal corrugated plate with the above geometric parameters was accordingly established with N corrugations. Eqs. (16) and (17) were used to calculate the equivalent elastic parameters and equivalent density of the trapezoidal corrugated plate as material parameters. The boundary conditions were first set as four simply supported edges, and a uniform load of 1000 N/m^2 was applied. The length of the corrugated plate in the corrugated direction was $a = 2cN$, and the length in the transverse direction was b. The quantitative relationship between corrugated and transverse directions is $a = b\gamma$, where γ is the aspect ratio of the plate. The equivalent stiffnesses of various corrugated plate geometries were substituted into Eq. (18) to obtain the theoretical mid-span deflection of each plate. The comparison of the

Table 7. Mid-span bending deflection of corrugated plate with four simply supported edges under uniform load.

N	γ	Finite element model ① (mm)	Xia et al. [31] ② (mm)	Present study ③ (mm)	Relative error (②-①)/①(%)	Relative error (③-①)/①(%)
8	0.5	0.0232	0.0199	0.0219	−14.073	−5.517
	1	0.3669	0.3393	0.3723	−7.524	1.450
	1.5	1.6781	1.5883	1.7150	−5.352	2.197
	2	4.1485	3.9759	4.2142	−4.159	1.585
12	0.5	0.1139	0.1009	0.1109	−11.387	−2.563
	1	1.8753	1.7178	1.8845	−8.395	0.495
	1.5	8.6272	8.0407	8.6820	−6.798	0.635
	2	21.2805	20.1281	21.3346	−5.415	0.254
16	0.5	0.3559	0.3189	0.3506	−10.415	−1.494
	1	5.9350	5.4292	5.9561	−8.522	0.355
	1.5	27.4155	25.4126	27.4394	−7.306	0.087
	2	67.5639	63.6148	67.4278	−5.845	−0.201
20	0.5	0.8645	0.7785	0.8560	−9.951	−0.984
	1	14.5449	13.2550	14.5412	−8.869	−0.025
	1.5	67.0999	62.0425	66.9906	−7.537	−0.163
	2	165.2960	155.3097	164.6187	−6.041	−0.410

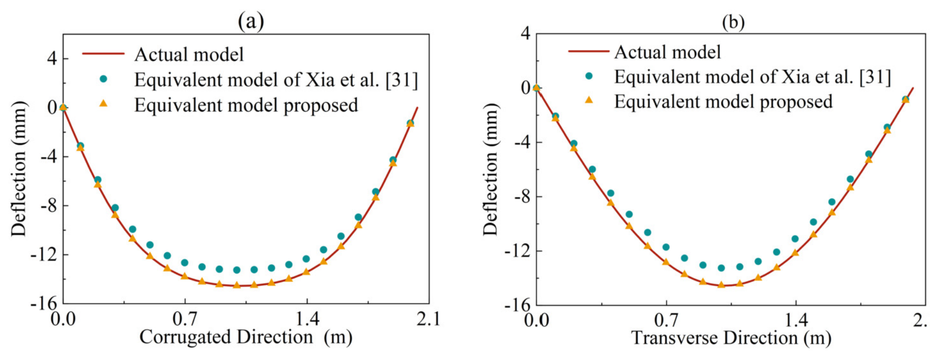

Fig. 3. Deflection at midway along trapezoidal corrugated plate with four simply supported edges in the (a) corrugated direction and (b) transverse direction.

theoretical and finite element-obtained mid-span deflections is shown in Table 7.

The theoretical mid-span deflection obtained using the equivalent model from [31] was consistently smaller than both the finite element and proposed model, indicating a larger equivalent transverse bending stiffness than the actual value, again confirming that the equivalent transverse bending stiffness obtained using the Dirichlet boundary condition is the upper limit of the real solution. Furthermore, the calculation precision was considerably improved with increasing aspect ratio γ. However, the computational accuracy of the equivalent model proposed in this study increased rapidly with increasing N, indicating that this equivalent model is more suitable for simulating a situation with more corrugations. Finally, the equivalent stiffnesses of the proposed model were closer to the real value (represented by the finite element model solution), indicating extremely high accuracy.

Three different finite element models representing the equivalent orthotropic plates from the proposed model and Xia et al. [31], as well as the actual trapezoidal corrugated plate, were then established for comparative analysis. Respectively, Figs. 3a and b show the deflection curves midway along the corrugated direction and in the transverse direction of the plate with four simply supported edges when the number of corrugations N was 20. It can be observed that the results obtained using the equivalent model proposed in this study coincided well with the finite element results for the trapezoidal corrugated plate.

The three plate finite element models were then evaluated under a uniform load of 1000 N/m^2 with two adjacent fixed edges and two free edges as the boundary conditions. Figs. 4a and b show the resulting deflection magnitude and rotation angle curves, respectively, of the free corrugated edge of the three plate models.

Next, the three plate finite element models were evaluated under a uniformly distributed load of 1000 N/m^2 with four fixed edges as the boundary conditions. Figs. 5a and b show the deflection magnitude and rotation angle curves, respectively, of the equivalent orthotropic plate and trapezoidal corrugated plate at mid-span in the corrugated direction.

As shown in Figs. 3–5, it can be observed that regardless of the boundary conditions, the results obtained using the equivalent model proposed in this study were close to the finite element simulation results. Indeed, the static response accuracy of the equivalent model proposed in this study was better than that of the equivalent model proposed by Xia et al. [31].

4. Dynamic verification of equivalent model

The bending and torsional stiffnesses obtained using the proposed equivalent model were used to predict the natural frequency of the corrugated plate [45]. This theoretical solution was compared with the results of a finite element analysis to verify the accuracy of the proposed equivalent stiffness model at high-order frequencies, as this was identified as a weakness of previous equivalent models.

The differential equation describing the mode shape of an orthotropic plate is given as follows:

$$D_1\frac{\partial^4 W}{\partial x^4}+D_2\frac{\partial^4 W}{\partial y^4}+2H\frac{\partial^4 W}{\partial x^4\partial y^4}-w^2\rho\delta W=0, \tag{19}$$

where W is the mode shape function. For the equivalent model, the equivalent density and thickness of the corrugated plate were used as follows:

$$\rho_{eq}=\rho,\quad h=\delta. \tag{20}$$

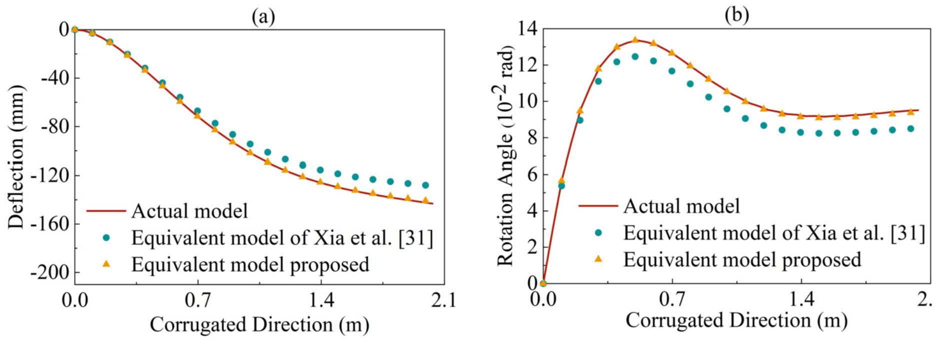

Fig. 4. Trapezoidal corrugated plate (a) deflection and (b) rotation angle at free corrugated edge with two fixed edges.

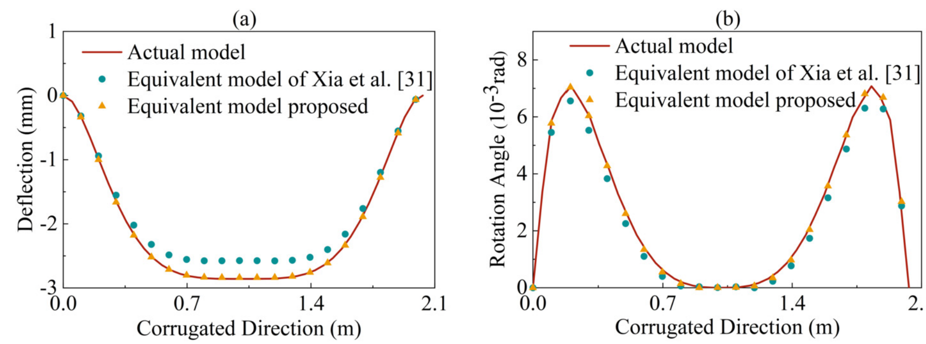

Fig. 5. Trapezoidal corrugated plate (a) deflection and (b) rotation angle at midspan in the corrugated direction with four fixed edges.

When the boundary condition of an orthotropic plate is simply supported on four edges, the mode shape function satisfying the boundary condition is

$$W(x,y)=\sin\left(\frac{m\pi x}{a}\right)\sin\left(\frac{n\pi y}{b}\right), \tag{21}$$

where m and n are positive integers as defined above.

Substituting Eqs. (20) and (21) into Eq. (19), the following expression can be obtained:

$$\left[D_1\left(\frac{m\pi}{a}\right)^4+D_2\left(\frac{n\pi}{b}\right)^4+2H\left(\frac{m\pi}{a}\right)^2\left(\frac{n\pi}{b}\right)^2-w^2\rho_{eq}h\right]$$
$$\times\sin\left(\frac{m\pi x}{a}\right)\sin\left(\frac{n\pi y}{b}\right)=0. \tag{22}$$

For all points on the central plane of the plate to satisfy this condition, Eq. (22) requires

$$D_1\left(\frac{m\pi}{a}\right)^4+D_2\left(\frac{n\pi}{b}\right)^4+2H\left(\frac{m\pi}{a}\right)^2\left(\frac{n\pi}{b}\right)^2-w^2\rho_{eq}h=0. \tag{23}$$

When the nonzero solution of W is obtained, Eq. (23) can be transformed into

$$w^2=\frac{D_1\left(\frac{m\pi}{a}\right)^4+D_2\left(\frac{n\pi}{b}\right)^4+2H\left(\frac{m\pi}{a}\right)^2\left(\frac{n\pi}{b}\right)^2}{\rho_{eq}h}. \tag{24}$$

When the values of m and n are different, the natural frequencies corresponding to the different modes can be obtained by

$$w_{mn}=\pi^2\sqrt{\frac{D_1\left(\frac{m}{a}\right)^4+D_2\left(\frac{n}{b}\right)^4+2H\left(\frac{m}{a}\right)^2\left(\frac{n}{b}\right)^2}{\rho_{eq}h}}. \tag{25}$$

A frequency analysis was then carried out using a finite element model of a trapezoidal corrugated plate with 4 simply supported edges and 20 corrugations, as described in Section 3.2.2. The theoretical natural frequency is calculated according to Eq. (25). The results of the finite element analysis are compared with those of the equivalent prediction model provided by Xia et al. [31] and the proposed equivalent prediction model in Table 8.

Compared with the finite element analysis results, the accuracies of the low-order natural frequencies predicted by the proposed equivalent model were notably better than those obtained using the Xia et al. model [31]. When predicting high-order natural frequencies, the transverse relative error of the proposed model was only 4.818%, showing improved accuracy in contrast to the model proposed by Xia et al. [31].

5. Application in the structure of corrugated metal pipe culvert

5.1. Arc-and-tangent corrugated plates

The arc-and-tangent corrugated plates comprise arc and tangent segments widely used in civil engineering, such as a corrugated steel shear wall, underground pipe gallery, and culvert. The section and section characteristic parameters of the arc-and-tangent corrugated plates are shown in Fig. 6(a), and the Cartesian coordinate system and local curvilinear coordinate system are shown in Fig. 6(b). Here, r is the inner curvature radius of the arc segment, TL is the length of the tangent segment, α_0 is the tangent angle, $\alpha=\alpha_0\times\pi/180°$, $2c$ represents the wave pitch, and $2f$ represents the wave depth.

The arc-and-tangent corrugated plate demonstrates symmetry within a single periodic waveform. $z(x)$ in the coordinates of any point on a quarter of a period can be expressed as a piecewise function as follows:

$$z(x)=\left\{\begin{array}{cc} x\tan\alpha & (0\le x\le X_1)\\ f-R+\sqrt{R^2-(X_2-x)^2} & (X_1\le x\le X_2)\end{array}\right\}, \tag{26}$$

where $R=r+h/2$, $X_1=TL/2\times\cos\alpha$, $X_2=X_1+R\sin\alpha$, $S_1=TL/2$, and $S_2=S_1+R\alpha$.

The corresponding half-length l and two integral expressions of the waveform are calculated as follows:

$$l=TL+2R\alpha,\quad I_1=R(2\alpha+\sin 2\alpha)+2TL\cos^2\alpha,$$
$$I_2=4\int_{S_1}^{S_2}\left(f-R+R\sqrt{1-\sin^2\left(\frac{\frac{TL}{2}-s+R\alpha}{R}\right)}\right)^2\mathrm{d}s+\frac{TL^3}{6}\sin^2\alpha. \tag{27}$$

Table 8. Natural frequency of trapezoidal corrugated plate with four simply supported edges.

Natural frequency	Finite element model ① (Hz)	Xia et al. [31] ② (Hz)	Present study ③ (Hz)	Relative error	
				(②-①)/①(%)	(③-①)/①(%)
w_{11}	40.983	42.802	40.986	4.249	0.005
w_{21}	50.799	52.151	50.671	2.594	−0.251
w_{31}	75.009	75.820	74.810	1.071	−0.265
w_{41}	114.530	115.078	114.415	0.476	−0.101
w_{12}	158.010	165.844	158.332	4.724	0.204
w_{22}	164.821	171.208	163.942	3.731	−0.533
w_{13}	351.934	371.392	354.407	5.239	0.703
w_{14}	620.100	659.224	628.979	5.935	1.432
w_{15}	959.317	1029.312	982.020	6.800	2.367
w_{16}	1365.776	1481.649	1413.521	7.821	3.496
w_{17}	1835.067	2016.232	1923.481	8.985	4.818

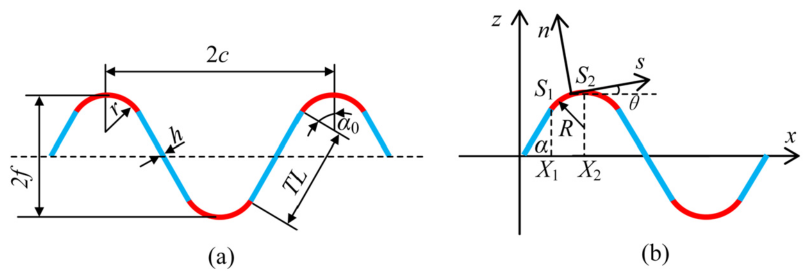

Fig. 6. Arc-and-tangent corrugated plate: (a) section characteristic parameters with (b) coordinate schematic.

The first term of I_2 does not have an explicit expression, and the numerical solution is used in the calculation.

5.2. Comparison of the values provided by the specification

Taking the 150×50 mm corrugated plate (i.e., wave pitch $2c$ = 150 mm and wave depth $2f = 50$ mm) given in Cold-formed corrugated steel pipes [50] as an example, the section area per unit length A_u, moment of inertia per unit length I_u, and plastic section modulus per unit length W_u were calculated. The formulas for A_u, I_u, and W_u are as follows:

$$A_u = \frac{A}{2c} = \frac{1}{2c}\int \mathrm{d}A = \frac{1}{2c}\int h\mathrm{d}s = \frac{h}{2c}\int_0^{2c}\sqrt{1+(z'(x))^2}\mathrm{d}x = \frac{hl}{c}, \tag{28}$$

$$I_u = \frac{I}{2c} = \frac{1}{2c}\int (z(x))^2\mathrm{d}A = \frac{1}{2c}\int (z(x))^2 h\mathrm{d}s$$
$$= \frac{h}{2c}\int_0^{2c}(z(x))^2\sqrt{1+(z'(x))^2}\mathrm{d}x = \frac{hI_2}{2c}, \tag{29}$$

$$W_u = \frac{I_u}{f+\frac{h}{2}} = \frac{hI_2}{2c\left(f+\frac{h}{2}\right)}. \tag{30}$$

According to Eqs. (26) and (27), the relative physical quantities and integral expressions of arc-and-tangent corrugated plates are calculated and substituted with the above results into Eqs. (28)-(30). In contrast to the values provided by the specification, the accuracy of the integral calculation of the section characteristic parameters of the arc-and-tangent corrugated plate in this study was verified, as shown in Table 9.

It can be observed from Table 9 that the proposed integral method for calculating the section characteristics of the corrugated steel plate has high accuracy, with a maximum error of 5.76%, which is acceptable in engineering.

5.3. Calculation example of buried corrugated metal pipe arch culvert

5.3.1. Full-scale test of pipe arch culvert

A full-scale test of a corrugated metal pipe arch culvert was performed in an open gravel pit in Enkoping, Sweden [51]. The inside span and inside height of the pipe arch were 6 and 4.55 m, respectively. The detailed dimensions of the culvert profile are shown in Fig. 7. The corrugation used for the steel plates was $200 \times 55 \times 3$ mm, and the pipe length was 5 m. The surrounding area including above the pipe was backfilled with 19 layers of uniformly graded soil. Crushed stone aggregate bedding material was used under the pipe. The backfilling process started 750 mm above the culvert bed. Compaction layers, each with an average thickness of 300 mm, were added until the cover depth reached 1.5 m. Symmetrical backfilling was used throughout the process.

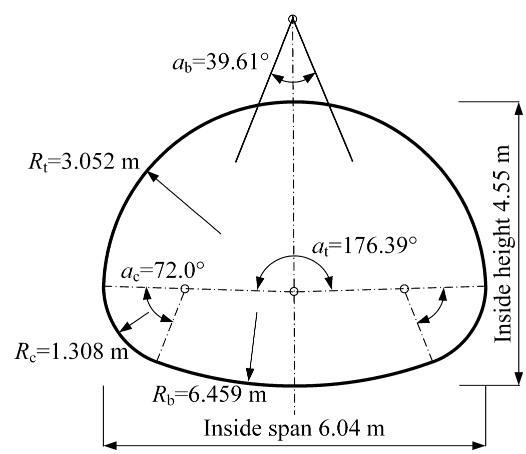

Fig. 7. Profile of corrugated metal pipe arch culvert.

5.3.2. Numerical modelling

The three-dimensional finite element programme PLAXIS 3D was used for numerical modelling and analysis of the pipe arch backfilling process. During the modelling process, three main components, i.e., the backfill soil, corrugated metal pipe, and model boundaries, were carefully simulated. The backfill soil was modelled as a set of layers of 300 mm thickness. The steel plates were modelled as orthotropic elements, and two equivalent orthotropic plate methods (one proposed in this study and the other in [46]) were used to examine the corrugation profile. Tables 10 and 11 show the properties of the actual corrugated plate

Table 9. Comparison with values in the specification.

h (mm)	A_u (mm²/mm)		error (%)	I_u (mm⁴/mm)		error (%)	W_u (mm³/mm)		error (%)
	Eq. (22)	[47]		Eq. (23)	[47]		Eq. (24)	[47]	
2	2.478	2.478	0.00	716.47	717.58	0.15	27.56	27.60	0.14
3	3.720	3.720	0.00	1080.99	1084.74	0.35	40.79	40.93	0.34
4	4.965	4.965	0.00	1449.74	1458.63	0.61	53.70	54.02	0.59
5	6.211	6.211	0.00	1822.85	1840.13	0.94	66.29	66.91	0.93
6	7.460	7.460	0.00	2200.14	2230.13	1.34	78.58	79.65	1.34
7	8.711	8.711	0.00	2581.78	2629.50	1.81	90.59	92.26	1.81
8	9.964	9.964	0.00	2967.96	3039.13	2.34	102.34	104.80	2.35
9	11.220	11.220	0.00	3358.37	3459.93	2.94	113.84	117.29	2.94
10	12.479	12.479	0.00	3753.24	3982.81	5.76	125.11	129.76	3.58

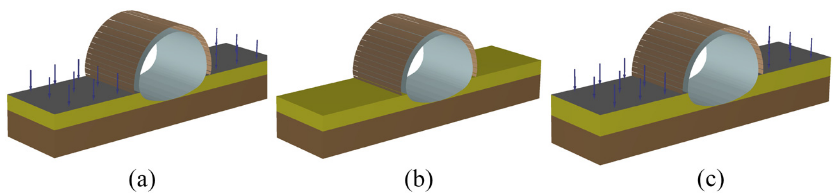

Fig. 8. Procedure to simulate compaction: (a) activate the soil layer and apply a surface load, (b) remove the surface load, and (c) repeat for the next soil layer.

Table 10. Actual (isotopic) material properties of the corrugated steel plate.

Parameter	Symbol	Value	Units
Plate thickness	h	3	mm
Inner curvature radius of the arc segment	r	53	mm
Length of the tangent segment	TL	32.3	mm
Tangent angle	α_0	45.187	°
Plate's half wave pitch	c	100	mm
Plate's half wave depth	f	27.5	mm
Poisson's ratio	ν	0.3	-
Unit weight	γ	76.93	kN/m³
Elastic modulus	E	200	GPa

and equivalent orthotropic plate, respectively. In addition, the soil–steel interaction behaviour was simulated by creating an interface boundary around the pipe body.

Soil is a highly nonlinear material, and its mechanical behaviour is stress-dependent. The hardening soil with small strains (HSs) model is an advanced soil constitutive model. Compared with the linear elastic, Mohr–Coulomb, and hardening soil models, the HSs model can provide an improved simulation of the interaction between the backfill soil and buried pipe under different load types, thus providing more reliable deformation results [52]. Therefore, the HSs model is suitable for simulating the mechanical behaviour of soil. Table 12 presents the soil properties used for the HSs model.

To eliminate boundary effects and provide accurate results, the length and width of the geometry model should be sufficiently large when setting the model boundaries. The pipe arch culvert test was conducted in an open large pit; hence, the geometry model was reasonably set at 20 m length, 5 m width, and 8 m depth. The model boundaries were fixed in horizontal directions, while the vertical direction was free. The mesh size was extremely fine, with approximately 350,000 elements, to provide accurate results.

As the formation of the soil–steel structure system depends on the surrounding soil, compaction work was required to densify the backfill layer. The compaction of the soil provided lateral support for the buried pipe, thereby increasing the system's ability to withstand higher service loads. To simulate backfill compaction, the method for applying temporary surface load was used in [52]. Figs. 8(a-c) illustrate how compaction loads are considered in the finite element model. Each compaction layer was applied in two stages. The first involved activating a new layer and applying a uniform vertical compaction load on the surface (the magnitude must be determined empirically). Generally, a surface load of 15–30 kPa is defined as the compaction load.

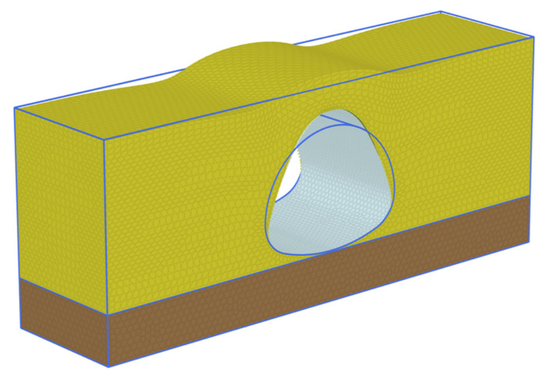

Fig. 9. Deformation mesh in numerical model for pipe arch full-scale test.

5.3.3. Comparison with experimental results

The pipe arch culvert belongs to the flexible culvert and will experience considerable vertical deflection during the backfilling process. The deformation of the pipe arch after backfilling is complete is shown in Fig. 9. The results were concentrated on the crown deflection and internal force of the pipe, which are expressed in terms of the thrust and bending moment. When the backfilling height reached the crown, the deflection and bending moment of the crown reached their maximum values simultaneously. When the backfilling height reached the cover height, the thrust of the crown reached its maximum value.

Fig. 10 shows the trend of crown deflection during the backfilling procedure. The maximum measured value in the experiment is 65 mm, while the results of this study and [46] are 64 mm and 60 mm, respectively.

Fig. 11 shows the change of crown thrust with increasing backfilling height. The measured value was −120 kN/m. The numerical simulation results of this study and [46] were approximately 112 and 106 kN/m, respectively.

Fig. 12 shows the change of the crown bending moment with the increasing backfilling height. In the process of backfilling, the maximum crown bending moment measured in the field was 10.3 kNm/m, and the calculated results of this study and [46] were extremely close to the measured value.

The two equivalent models could predict the deformation and internal forces of corrugated metal culverts well. The model proposed in this study was closer to the experimental values in the pipe crown deflection and thrust.

Table 11. Equivalent (orthotropic) material properties of the corrugated steel plate.

Parameter	Symbol	Value		Units
		Aagah and Aryannejad [46]	Present study	
Orthotropic plate thickness	$\overline{h}$	67.69	3	mm
Circumferential orthotropic elastic modulus	E_θ	10994	12042×10^4	MPa
Longitudinal orthotropic elastic modulus	E_L	18.28	18.60×10^4	MPa
Circumferential orthotropic out-of-plane shear modulus	$G_{\theta R}$	4228	6018×10^4	MPa
Longitudinal orthotropic out-of-plane shear modulus	G_{LR}	7.03	7.15×10^4	MPa
Orthotropic in-plane shear modulus	$G_{\theta L}$	1.28	9.09×10^4	MPa
Orthotropic Poisson's ratio	$v_{L\theta}$	0	4.63×10^4	-
Orthotropic unit weight	$\overline{\gamma}$	4.08	95.29	kN/m^3

Table 12. Characteristics of backfill in the small strain hardened soil (HSs) model.

Characteristics	Symbol	Value	Units
Unit weight of soil	γ_s	17.8	kN/m^3
Secant modulus	E_{50} *	16.4	MPa
Oedometer modulus	E_{oed} *	13.2	MPa
Unloading-reloading modulus	E_{ur} *	49.2	MPa
Poisson's ratio	υ'	0.2	-
Cohesion	c'	1	kPa
Friction angle	φ	36	degree
Dilation angle	ψ	6	degree
Interface factor	R_{in}	0.67	-
Shear modulus	G_o	150	MPa
Shear strain	$\gamma_{0.7}$	0.002	-
Earth pressure coefficient	K	0.41	-

* The value of E_{50} was chosen to be equal to the utilized backfill soil stiffness reported from field tests, while the values of E_{oed} and E_{ur} were chosen based on the range recommended for cohesionless soils.

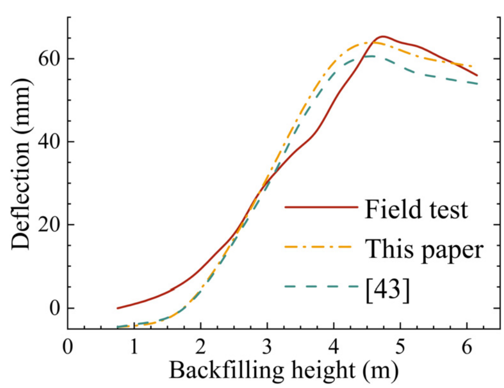

Fig. 10. Backfilling height versus crown deflections in the pipe arch; comparing field measurements with the results of this study and [43].

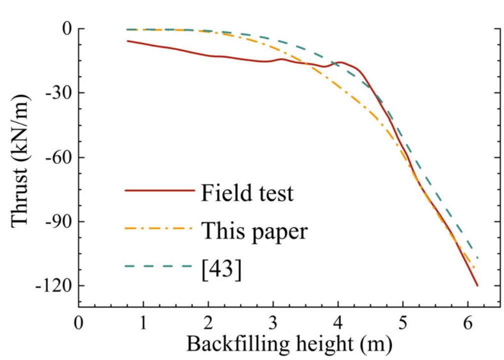

Fig. 11. Thrusts in the pipe arch during backfilling; comparing field measurements with the results of this study and [43].

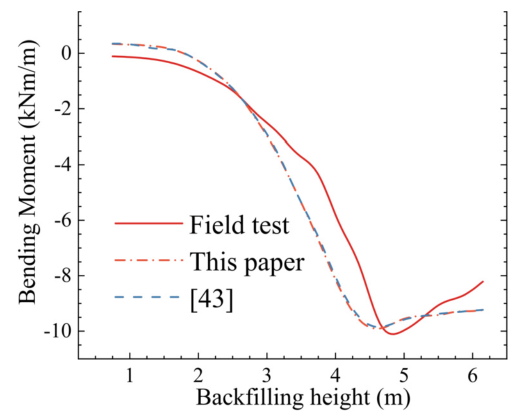

Fig. 12. Bending moments in the pipe arch during backfilling; comparing field measurements with the results of this study and [43].

6. Conclusions

Based on the Kirchhoff hypothesis, a new representative volume element method-based equivalent model of a corrugated plate was derived by reasonably simplifying the constitutive relation according to the generalised strain boundary conditions. Compared with the equivalent model provided by Xia et al. [31], the equivalent transverse bending stiffness obtained using the proposed equivalent model was found to be more consistent with the simplified expression in [33] based on the variational asymptotic method [34, 35]. It was found that the equivalent transverse bending stiffness of a corrugated plate of an isotropic material determined by [31] was consistently larger because the coefficient $1/(1-\nu^2)$ was always greater than 1. A subsequent analysis demonstrated that the simplified constitutive relation was reasonable under the generalised strain boundary conditions and could be applied to effectively improve the accuracy of the corrugated plate equivalent model.

The static response accuracy of the proposed corrugated plate equivalent model was affected by the aspect ratio of the corrugated plate and the number of corrugations because the representative volume element method is affected by the relative size of the corrugated plate and the single corrugation waveform. Therefore, when the size of the corrugated plate was much larger than the size of a single waveform, the response accuracy of the proposed equivalent model was extremely high. Compared with existing equivalent models, the new equivalent model provided a more accurate prediction of static response under different waveforms and boundary conditions. In terms of dynamic characteristics, the proposed equivalent model was shown to accurately predict the low-order natural frequencies of the corrugated plate and provide improved accuracy when predicting high-order natural frequencies.

The expressions of half-length l and two integrals relating to the equivalent stiffness of arc-and-tangent corrugated plates were provided. The proposed equivalent model was used to simulate a full-scale test of a corrugated metal pipe arch culvert. The results were in good agreement with the measured values.

Declarations

Author contribution statement

Kun Lang: Conceived and designed the experiments; Performed the experiments; Analyzed and interpreted the data; Contributed reagents, materials, analysis tools or data; Wrote the paper.

Mingzhou Su: Conceived and designed the experiments; Analyzed and interpreted the data.

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Data availability statement

No data was used for the research described in the article.

Declaration of interests statement

The authors declare no conflict of interest.

Additional information

No additional information is available for this paper.

References

[1] D. Twede, S.E. Selke, D.P. Kamdem, D. Shires, Cartons, Crates and Corrugated Board: Handbook of Paper and Wood Packaging Technology, DEStech Publications Inc., 2005.
[2] D. Dubina, V. Ungureanu, L. Gîlia, Cold-formed steel beams with corrugated web and discrete web-to-flange fasteners, Steel Constr. 6 (2) (2013) 74–81.
[3] J. Noll, S. Tysl, M. Westrich, The Use of Corrugated Metal Pipe and Structural Plate for Aggregate Tunnel and Conveyor Enclosure Applications, Professional Development Advertising Section, CONTECH Construction Products Inc., 2009.
[4] E.M. Knox, M.J. Cowling, I.E. Winkle, Adhesively bonded steel corrugated core sandwich construction for marine applications, Mar. Struct. 11 (4–5) (1998) 185–204.
[5] N. Tsushima, T. Yokozeki, W. Su, H. Arizono, Geometrically nonlinear static aeroelastic analysis of composite morphing wing with corrugated structures, Aerosp. Sci. Technol. 88 (2019) 244–257.
[6] V.N.V. Hoang, D.G. Ninh, D.V. Truong, H.V. Bao, V.L. Huy, Behaviors of dynamics and stability standard of graphene nanoplatelet reinforced polymer corrugated plates resting on the nonlinear elastic foundations, Compos. Struct. 260 (2021) 113253.
[7] L. Tian, H. Zhao, M. Yuan, G. Wang, B. Zhang, J. Chen, J. Chen, Global buckling and multiscale responses of fiber-reinforced composite cylindrical shells with trapezoidal corrugated cores, Compos. Struct. 260 (2021) 113270.
[8] M.T. Huber, Die Theorie des kreuzweise Bewehrten Eisenbetonplatten, Bauingenieur 4 (12) (1923) 354–360.
[9] E. Seydel, Shear buckling of corrugated plates, Jahrb. Dtsch. Versuchsanstalt Luftfahrt 9 (1931) 233–245.
[10] S.P. Timoshenko, S. Woinowsky-Krieger, Theory of Plates and Shells, McGraw-Hill, New York, 1959.
[11] D.E. McFarland, An investigation of the static stability of corrugated rectangular plates loaded in pure shear, Ph.D. Dissertation, University of Kansas, 1967.
[12] D. Briassoulis, Equivalent orthotropic properties of corrugated sheets, Comput. Struct. 23 (2) (1986) 129–138.
[13] A. Samanta, M. Mukhopadhyay, Finite element static and dynamic analyses of folded plates, Eng. Struct. 21 (3) (1999) 277–287.
[14] T. Yokozeki, S.I. Takeda, T. Ogasawara, T. Ishikawa, Mechanical properties of corrugated composites for candidate materials of flexible wing structures, Compos. A 37 (10) (2006) 1578–1586.
[15] I. Dayyani, S. Ziaei-Rad, H. Salehi, Numerical and experimental investigations on mechanical behavior of composite corrugated core, Appl. Compos. Mater. 19 (3–4) (2012) 705–721.
[16] G. Bartolozzi, M. Pierini, U.L.F. Orrenius, N. Baldanzini, An equivalent material formulation for sinusoidal corrugated cores of structural sandwich panels, Compos. Struct. 100 (2013) 173–185.
[17] G. Kress, M. Winkler, Corrugated laminate homogenization model, Compos. Struct. 92 (3) (2010) 795–810.
[18] G. Kress, M. Winkler, Corrugated laminate analysis: a generalized plane-strain problem, Compos. Struct. 93 (5) (2011) 1493–1504.
[19] D.T. Filipovic, G.R. Kress, A planar finite element formulation for corrugated laminates under transverse shear loading, Compos. Struct. 201 (2018) 958–967.
[20] D.T. Filipovic, G.R. Kress, Stress analysis of corrugated orthotropic laminates under transverse shear loading, Compos. Struct. 223 (2019) 110983.
[21] H. Mohammadi, S. Ziaei-Rad, I. Dayyani, An equivalent model for trapezoidal corrugated cores based on homogenization method, Compos. Struct. 131 (2015) 160–170.
[22] K.M. Liew, L.X. Peng, S. Kitipornchai, Nonlinear analysis of corrugated plates using a FSDT and a meshfree method, Comput. Methods Appl. Mech. Eng. 196 (21–24) (2007) 2358–2376.
[23] B.D. Annin, A.G. Kolpakov, S.I. Rakin, Homogenization of corrugated plates based on the dimension reduction for the periodicity cell problem, in: H. Altenbach, R. Goldstein, E. Murashkin (Eds.), Mechanics for Materials and Technologies, in: Advanced Structured Materials, vol. 46, Springer, Cham, 2017, pp. 49–72.
[24] N.S. Bakhvalov, Panasenko GP. Homogenization, Averaging Process in Periodic Structures, Kluwer Academic Publications, Dordrecht, 1989, pp. 55–67.
[25] A. Bensoussan, J.L. Lions, G. Papanicolaou, Asymptotic Analysis for Periodic Structures, North-Holland Publications, Amsterdam, 1978, pp. 34–58.
[26] E. Sanchez-Palencia, A. Zaoui, Homogenization Techniques for Composite Media, Springer Verlag, Berlin, 1987, pp. 29–41.
[27] B. Hassani, E. Hinton, A review of homogenization and topology optimization I—homogenization theory for media with periodic structure, Comput. Struct. 69 (6) (1998) 707–717.
[28] B. Hassani, E. Hinton, A review of homogenization and topology optimization II—analytical and numerical solution of homogenization equations, Comput. Struct. 69 (6) (1998) 719–738.
[29] B. Hassani, E. Hinton, A review of homogenization and topology optimization III—topology optimization using optimality criteria, Comput. Struct. 69 (6) (1998) 739–756.
[30] B. Hassani, E. Hinton, Homogenization and Structural Topology Optimization: Theory, Practice, and Software, Springer Science and Business Media, London, 1999, pp. 77–91.
[31] Y. Xia, M.I. Friswell, E.IS. Flores, Equivalent models of corrugated panels, Int. J. Solids Struct. 49 (13) (2012) 1453–1462.
[32] Y. Xia, M.I. Friswell, Equivalent models of corrugated laminates for morphing skins, in: Active and Passive Smart Structures and Integrated Systems 2011, International Society for Optics and Photonics, 2011, p. 7977, 79771I.
[33] J.C. Michel, H. Moulinec, P. Suquet, Effective properties of composite materials with periodic microstructure: a computational approach, Comput. Methods Appl. Mech. Eng. 172 (1–4) (1999) 109–143.
[34] C. Miehe, J. Schotte, M. Lambrecht, Homogenization of inelastic solid materials at finite strains based on incremental minimization principles. Application to the texture analysis of polycrystals, J. Mech. Phys. Solids 50 (10) (2002) 2123–2167.
[35] Z. Ye, V.L. Berdichevsky, W. Yu, An equivalent classical plate model of corrugated structures, Int. J. Solids Struct. 51 (11–12) (2014) 2073–2083.
[36] A.A. Kolpakov, A.G. Kolpakov, Computation of homogenized stiffnesses of thin corrugated plates, arXiv preprint, arXiv:1811.01718, 2018.
[37] A.A. Kolpakov, A.G. Kolpakov, On the effective stiffnesses of corrugated plates of various geometries, Int. J. Eng. Sci. 154 (2020) 103327.
[38] W. Yu, A unified theory for constitutive modeling of composites, J. Mech. Mater. Struct. 11 (4) (2016) 379–411.
[39] W. Yu, Simplified formulation of mechanics of structure genome, AIAA J. 57 (10) (2019) 4201–4209.
[40] A. Deo, W. Yu, Equivalent plate properties of composite corrugated structures using mechanics of structure genome, Int. J. Solids Struct. 208–209 (2021) 262–271.
[41] R. Hill, The elastic behaviour of a crystalline aggregate, Proc. Phys. Soc. A 65 (5) (1952) 349–354.
[42] J. Yan, G. Cheng, S. Liu, L. Liu, Comparison of prediction on effective elastic property and shape optimization of truss material with periodic microstructure, Int. J. Mech. Sci. 48 (4) (2006) 400–413.
[43] W. Voigt, On the relation between the elasticity constants of isotropic bodies, Ann. Phys. 274 (12) (1889) 573–587.
[44] A. Reuss, Determination of the yield point of polycrystals based on the yield condition of single crystals, Z. Angew. Math. Mech. 9 (1) (1929) 49–58.
[45] Y. Aoki, W. Maysenhölder, Experimental and numerical assessment of the equivalent-orthotropic-thin-plate model for bending of corrugated panels, Int. J. Solids Struct. 108 (2017) 11–23.
[46] O. Aagah, S. Aryannejad, Dynamic Analysis of Soil-Steel Composite Railway Bridges. Master of Science, KTH Royal Institute of Technology, 2014.
[47] K.M. El-Sawy, Three-dimensional modeling of soil-steel culverts under the effect of truckloads, Thin-Walled Struct. 41 (8) (2003) 747–768.
[48] C.Y. Lee, W. Yu, Homogenization and dimensional reduction of composite plates with in-plane heterogeneity, Int. J. Solids Struct. 48 (10) (2011) 1474–1484.
[49] R.M. Jones, Mechanics of Composite Materials, Hemisphere Publishing Corporation, New York, 1975.
[50] GB/T 34567–2017, Cold-Formed Corrugated Steel Pipes, Standards Press of China, Beijing (China), 2017 (in Chinese).
[51] L.G. Pettersson, Full scale tests and structural evaluation of soil steel flexible culverts with low height of cover, Doctoral Thesis, KTH Royal Institute of Technology, 2007.
[52] I. Ezzeldin, H.E. El Naggar, Three-dimensional finite element modeling of corrugated metal pipes, Transp. Geotech. 27 (7) (2021) 100467.